讨论1

- **应变片的主要特性**
  - 应变片电阻值：60Ω、120Ω、200Ω、350Ω、1000Ω
  - 灵敏系数 $K=\frac{\Delta R/R}{\Delta l/l}$
  - 绝缘电阻：应变计的敏感栅和引线与被测件之间的电阻
  - 横向效应
  - 机械滞后
  - 零漂和蠕变
    - 零漂：不承受机械应变时，指示应变值随时间变化
    - 蠕变：承受恒定的应变时，指示应变值随时间变化
  - 允许电流：应变片不因电流放热而影响测量所允许的最大电流
  - 应变极限 应变极限 ε_{lim}，$\varepsilon \le 10\%$时的最大真实应变 $\delta=\frac{|\varepsilon_s-\varepsilon_z|}{\varepsilon_s}\times 100\%$

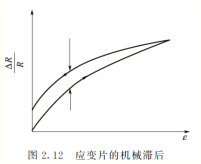


图 2.12 应变片的机械滞后

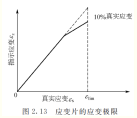


图 2.13 应变片的应变极限

1

电阻应变片的温度补偿

- 应变片电阻具有电阻温度系数
- 弹性元件与应变片电阻两者的线膨胀系数不同，即使无外力作用，即无应变现象，由于环境温度的变化也会引起应变片电阻值的改变，从而产生测量误差。



2

讨论：

- **电桥工作原理？**

3

直流电桥&交流电桥

作用：将电阻的变化转换为电压输出。

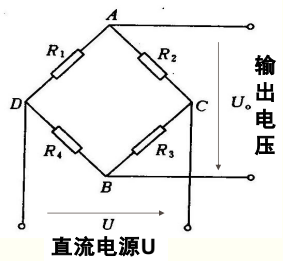
◆ 直流电桥

空载电压：

$$U_0=U_{AB}=U_{AC}-U_{BC}=\frac{U}{R_1+R_2}R_2-\frac{U}{R_3+R_4}R_3$$

$$U_0=\frac{R_1R_3-R_2R_4}{(R_1+R_2)(R_3+R_4)}U$$

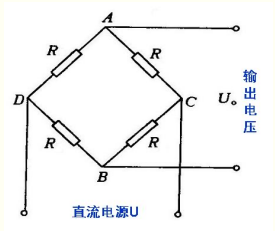
等效电阻： $R'=\frac{R_1R_2}{R_1+R_2}+\frac{R_3R_4}{R_3+R_4}$



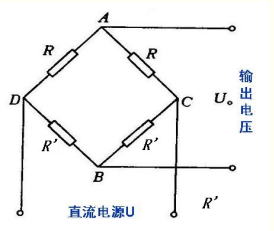
直流电桥平衡条件：

$$R_1R_3=R_2R_4$$

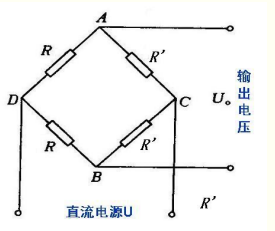
4



等臂电桥



输出对称电桥



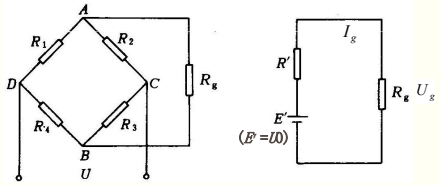
电源对称电桥

5

直流电桥的电流输出

如果直流电桥带一个具有内阻R_g的负载,将输出电流I_g.

电流输出电路及其等效电路:



6

由等效电路可求得I_g、U_g的一般表达式:

$$I_g=\frac{U_0}{R'+R_g}=U\frac{R_1R_3-R_2R_4}{R_g(R_1+R_2)(R_3+R_4)+R_1R_2(R_3+R_4)+R_3R_4(R_1+R_2)}$$

$$U_g=I_gR_g=R_g\frac{R_1R_3-R_2R_4}{R_g(R_1+R_2)(R_3+R_4)+R_1R_2(R_3+R_4)+R_3R_4(R_1+R_2)}$$

当阻抗匹配即R_g=R'时:

$$I_g=\frac{U}{2}\frac{R_1R_3-R_2R_4}{R_1R_2(R_3+R_4)+R_3R_4(R_1+R_2)}$$

$$U_g=\frac{U}{2}\frac{R_1R_3-R_2R_4}{(R_1+R_2)(R_3+R_4)}$$

7

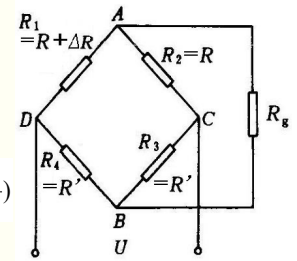
(1) 阻抗匹配的输出对称电桥

设R_1为应变电阻，当电阻增加ΔR时:

$$I_g=\frac{U}{2}\frac{R_1R_3-R_2R_4}{R_1R_2(R_3+R_4)+R_3R_4(R_1+R_2)}$$

得输出电流为: $\Delta I_g=\frac{U}{2}\frac{\Delta RR'}{R^2 2R'+R'^2 2R}=\frac{U}{4}\frac{1}{R+R'}(\frac{\Delta R}{R})$

即 $\Delta I_g=\frac{U}{4}\frac{K\varepsilon}{R+R'}$



(2) 阻抗匹配的电源对称电桥，也有 $\Delta I_g=\frac{U}{4}\frac{K\varepsilon}{R+R'}$

(3) 阻抗匹配的等臂电桥，有 $\Delta I_g=\frac{U}{8R}K\varepsilon$

8

直流电桥的电压输出

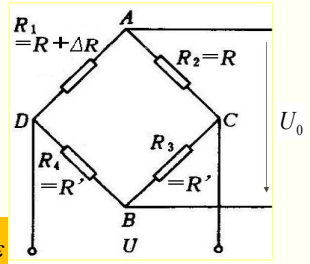
如果直流电桥输出带运算放大器，负载电阻$R_g\approx\infty$,输出电流$I_g\approx0$，输出电压即可认为是电桥电路的开路电压：

对于输出对称电桥

$$U_0=\frac{R_1R_3-R_2R_4}{(R_1+R_2)(R_3+R_4)}U$$

$$U_0=\frac{\Delta RR'}{(R+R)(R'+R')}U=\frac{U}{4}\frac{\Delta R}{R}$$

即 $U_0=\frac{U}{4}K\varepsilon$



对于电源对称电桥

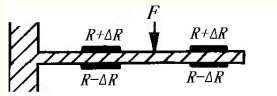
$$U_0=U\frac{RR'}{(R+R')^2}K\varepsilon$$

对于等臂电桥

$$U_0=\frac{U}{4}K\varepsilon$$

9

直流电桥实用电路----等臂电桥全桥工作电路

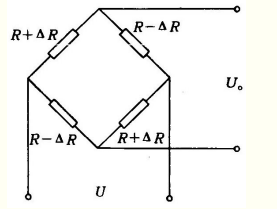


特征：四个电阻相同

四个电阻全为应变电阻

（两个被拉长，两个被压短）

$$U_0=\frac{R_1R_3-R_2R_4}{(R_1+R_2)(R_3+R_4)}U=\frac{(R+\Delta R)^2-(R-\Delta R)^2}{(2R)(2R)}U=\frac{4R\Delta R}{4R^2}U=U\frac{\Delta R}{R}$$



$U_0=UK\varepsilon$ **Vs.** $U_0=\frac{U}{4}K\varepsilon$

与单臂应变相比，灵敏度扩大4倍！

10

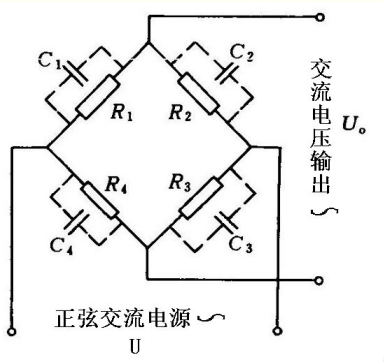
- **交流电桥**

交流电桥采用正弦交流电源U，电源频率取被测应变频率的5~10倍。各桥臂采用应变片或定值精密无感电阻。

在频率较高时，必须考虑分布电容的影响。

如：

$$Z_1=R_1//\frac{1}{j\omega C_1}$$



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设正弦交流电源 $u=U_m\sin\omega t$

则输出交流电压 $u_0(t)=\frac{Z_1Z_3-Z_2Z_4}{(Z_1+Z_2)(Z_3+Z_4)}U_m\sin\omega t$

或 $\dot{U}_0=\frac{Z_1Z_3-Z_2Z_4}{(Z_1+Z_2)(Z_3+Z_4)}\dot{U}$

交流电桥平衡条件：

$$Z_1Z_3=Z_2Z_4$$

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等臂电桥单臂应变情况下

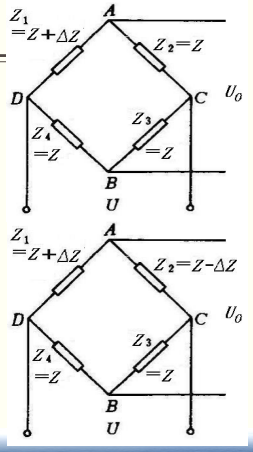
$$\dot{U}_0 = \frac{1}{4}\frac{\Delta Z}{Z}\dot{U}$$

$$u_0(t) = \frac{1}{4}K\varepsilon U_m \sin \omega t$$

等臂电桥双臂应变情况下

$$\dot{U}_0 = \frac{1}{2}\frac{\Delta Z}{Z}\dot{U}$$

$$u_0(t) = \frac{1}{2}K\varepsilon U_m \sin \omega t$$

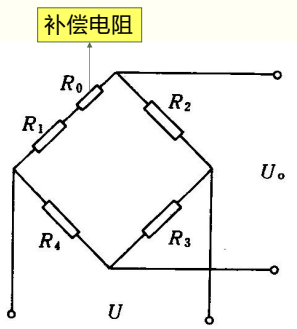


13

- **电桥的线路补偿**
  - 初始平衡补偿

对平衡位置（零点）的改变进行的补偿。

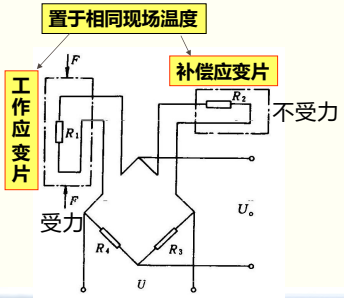
如果某一对桥臂电阻的乘积较小，可对这两个电阻中的任一个串联小的补偿电阻进行补偿，以满足电桥的初始平衡条件。



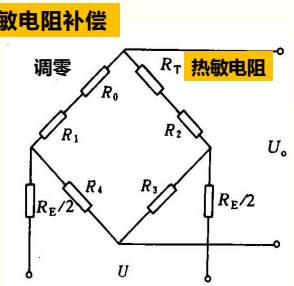
14

- **温度补偿**

a.补偿片补偿



b.热敏电阻补偿



如图中R2、R4桥臂的总温度系数低，可采用高温度系数的R_T进行补偿

15

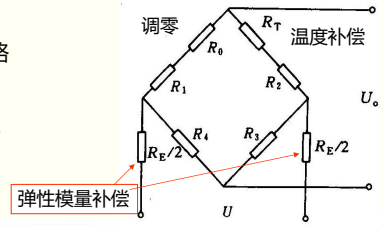
- **弹性模量补偿**

弹性材料受环境影响，如升温变软，弹性模量减小；老化变硬，弹性模量增大等。

对于升温变软，可采用铜丝或镍丝制成补偿电阻R_E进行补偿。

分析：当弹性材料升温变软，同样受力引起的应变ε增大，测量电路输出会增大。

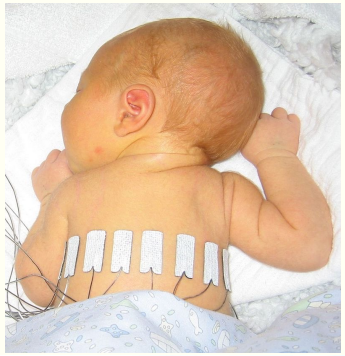
加上R_E之后，由于R_E升温增大，电桥电路供电电压减小，补偿了升温引起的输出增大。



16

讨论2

- **What !?**
- **How?**



17

CT (Computed Tomography)



Tomography is a technology that involves taking measurements around the periphery of an object (e.g. process vessel or pipe) to determine what is going on inside.



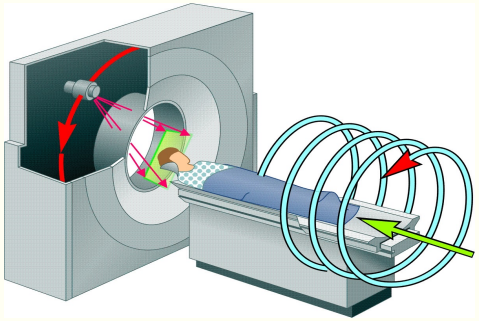
18

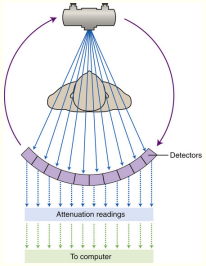
- **Tomography**
  - tomo: Greek tomos[section]

    希腊语 tomos[切面]

  - graphy: Latin *-graphia,* from Greek, from *graphein.* writing or representation in a (specified) manner or by a (specified) means or of a (specified) object.

19





20

History of CT

- 1895, Wilhelm Conrad Röntgen discovers x-rays in Würzburg, Bavaria, Germany.



Herr Professor Dr & Mrs Röntgen, December 1895

- 1917, Johann Radon demonstrates that the image of a 3-dimensional object can be reconstructed from an infinite number of 2-dimensional projections of the object, providing the mathematical basis for CT image construction (Vienna, Austria).

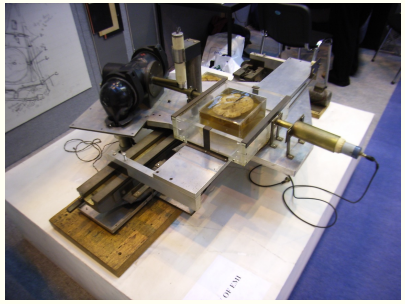
From impactscan.org | a brief history of CT

21

- 1963-1964, Allan Cormack publishes in the Journal of Applied Physics a theoretical analysis and the results from experimental scanner using a computer to reconstruct cross-sectional images from data (Medford, Massachusetts, USA).

22

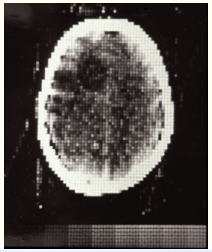
- 1968, Godfrey Hounsfield's original project proposal EMI





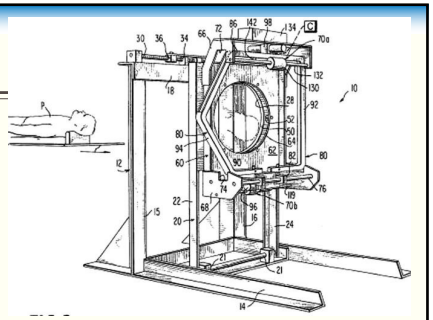
23

- 1971, First patient (head scan), under the supervision of Hounsfield and Ambrose, at Atkinson Morley's Hospital, using prototype EMI head scanner.



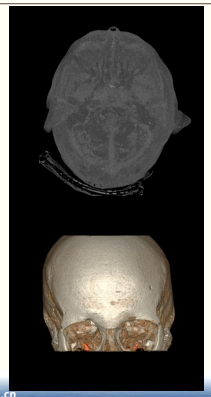
24

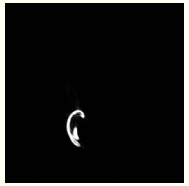
- 1972, Godfrey Hounsfield & James Bull lecture, New York, showing first clinical CT images; and The first CT scanner demonstration in the United States, at the Mayo Clinic.
- 1973, first clinical patient scan in USA, at the Mayo Clinic; Robert Ledley designs ACTA, a whole body CT scanner, later marketed by Pfizer





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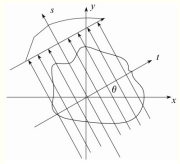
26

- **层析成像(Tomography) 也称为计算机层析(断层) 成像(Computerized Tomography, CT)或计算机辅助层析成像(Computer Assisted Tomography, CAT), 是指在不损伤研究对象内部结构的条件下, 利用某种探测源, 根据从对象外部设备所获得的投影数据, 运用一定的数学模型和重建技术, 使用计算机生成对象内部的二维/三维图像,重现对象内部特征。层析成像不同于从“图像到图像”的常规计算机图像处理技术, 而是由投影数据重建反映对象内部特征的图像, 是一类特殊的图像处理技术, 常称为“图像重建”。**

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数学原理：Radon Transform

$$Rf(t,\theta)=\int_{-\infty}^{\infty} f(t\cos\theta-s\sin\theta,t\sin\theta+s\cos\theta)\mathrm{d}s$$





$$f(x,y)=-\frac{1}{2\pi^2}\lim_{\varepsilon\to 0}\int_{\varepsilon}^{\infty}\frac{1}{q}\int_{0}^{2\pi} Rf_1(x\cos\theta+y\sin\theta+q,\theta)\mathrm{d}\theta\mathrm{d}q$$

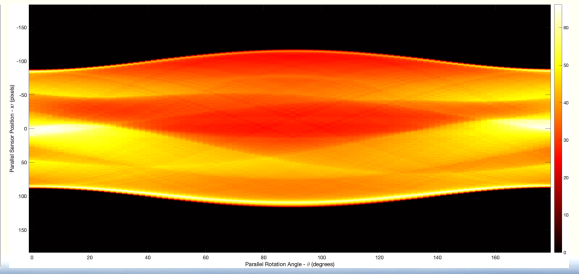
28

```
P = phantom(256);
Imshow(P)
```

```
theta3 = 0:2:178;
[R3,xp] = radon(P,theta3);
```

```
imagesc(theta3,xp,R3)
colormap(hot)
colorbar
xlabel('Parallel Rotation Angle - \theta (degrees)');
ylabel('Parallel Sensor Position - x\prime (pixels)');
```

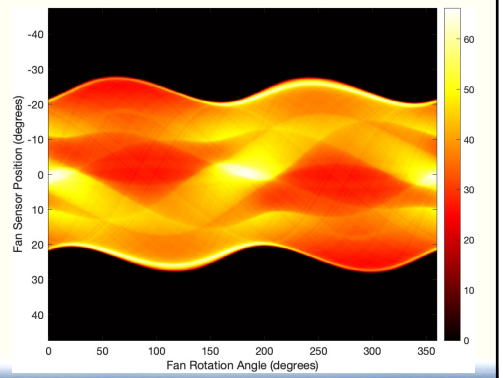




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```
dsensor3 = 0.25;
[F3, sensor_pos3, fan_rot_angles3] =
fanbeam(P,D,... 'FanSensorSpacing',dsensor3);

imagesc(fan_rot_angles3, sensor_pos3, F3)
colormap(hot)
colorbar
xlabel('Fan Rotation Angle (degrees)')
ylabel('Fan Sensor Position (degrees)')
```



30

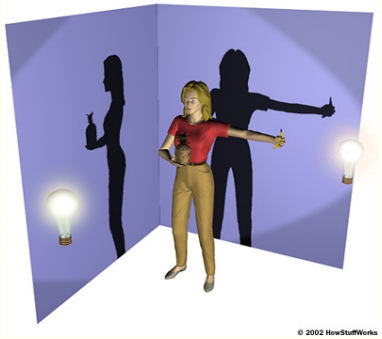
- 投影（Projection）







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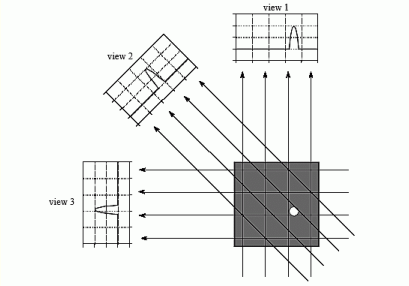


FIGURE 25-15
CT views. Computed tomography acquires a set of views and then reconstructs the corresponding image. Each sample in a view is equal to the sum of the image values along the ray that points to that sample. In this example, the image is a small pillbox surrounded by zeros. While only three views are shown here, a typical CT scan uses hundreds of views at slightly different angles.

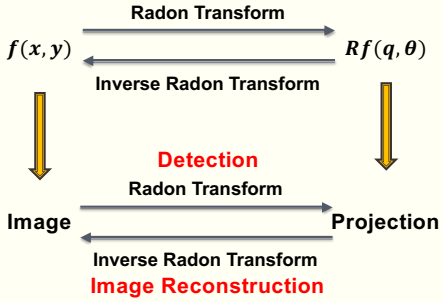
33

一般将函数$f(x,y)$称为“图像”(Image)，将$Rf(q,\theta)$（线积分）称为该“图像”沿某一投影方向（直线）上的“投影”（Projection），则Radon逆变换的实质就是：由“图像”在所有方向上的“投影”可“重建”该“图像”。

相应的实现Radon逆变换的过程，即完成投影重建图像的过程，一般称为“图像重建”（Image Reconstruction）。

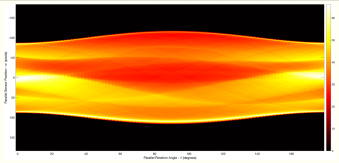
$$f(x,y)=-\frac{1}{2\pi^2}\lim_{\varepsilon\to 0}\int_{\varepsilon}^{\infty}\frac{1}{q}\int_{0}^{2\pi}Rf_1(x\cos\theta+y\sin\theta+q,\theta)\mathrm{d}\theta\mathrm{d}q$$

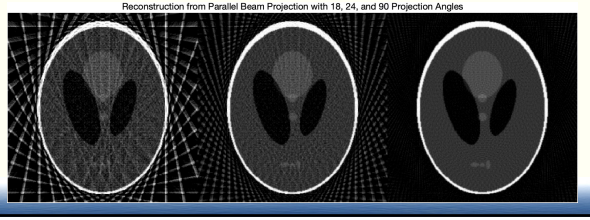
34



35

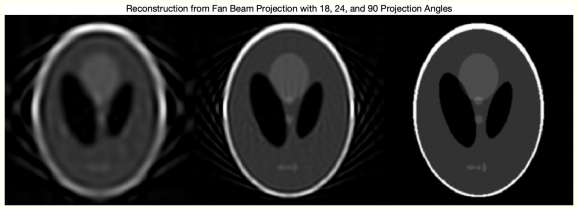
```
utput_size = max(size(P));
dtheta1 = theta1(2) - theta1(1);
I1 = iradon(R1,dtheta1,output_size);
dtheta2 = theta2(2) - theta2(1);
I2 = iradon(R2,dtheta2,output_size);
dtheta3 = theta3(2) - theta3(1);
I3 = iradon(R3,dtheta3,output_size);
figure
montage({I1,I2,I3},'Size',[1 3])
title('Reconstruction from Parallel Beam Projection with 18, 24, and 90 Projection Angles')
```





36

```
Ifan1 = ifanbeam(F1,D,'FanSensorSpacing',dsensor1,'OutputSize',output_size);
Ifan2 = ifanbeam(F2,D,'FanSensorSpacing',dsensor2,'OutputSize',output_size);
Ifan3 = ifanbeam(F3,D,'FanSensorSpacing',dsensor3,'OutputSize',output_size);
figure
montage({Ifan1,Ifan2,Ifan3},'Size',[1 3])
title('Reconstruction from Fan Beam Projection with 18, 24, and 90 Projection Angles')
```



37

过程层析成像(Process Tomography, PT)技术也常称为流动成像(Flow Imaging)技术，是20世纪80年代中后期开始正式形成和发展起来的，以CT技术为基础的，一种以两相流或多相流为主要对象的过程参数二维或三维分布状况的在线实时检测技术。过程层析成像是CT技术与工业要求相结合的产物。

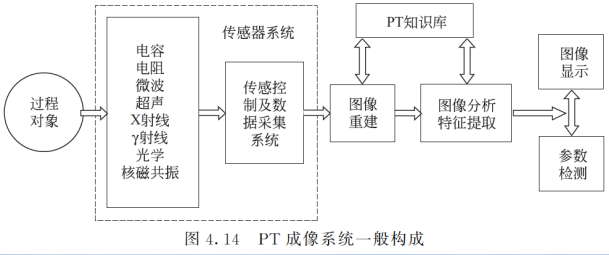


图 4.14 PT 成像系统一般构成

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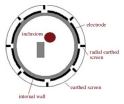
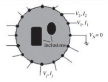
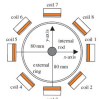
- 如何获得投影数据?
- 如何进行图像重建?

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如何获得投影数据?

tomography technique	characteristics	applications
electromagnetism (hard field)	gamma-ray	radioactive source, fast scanning	industrial and medical imaging
	X-ray	high resolution, mechanically scanned, radiation confinement	industrial and medical imaging
	UV, optical	fast, optical access	microstructure and medical monitoring
	millimetre wave	system component emerging	air traffic monitoring
electromagnetism (soft field)	microwave	fast, moderate resolution (wavelength dependence)	thermal mapping of reactor, breast cancer imaging
	capacitive (ECT)	low resolution, contactless	passive electrical property imaging, in processes and materials
	impedance (EIT)	fast, low cost	medical, industrial and geophysical applications
	inductance (MIT)	contactless, low cost	medical and industrial imaging
nuclear particle	positron emission	similar to computed tomography reconstruction, functional imaging	medical imaging
	neutron	high resolution, radioactive source	microstructure, industrial imaging
others	MRI	high resolution, soft-tissue imaging	medical and industrial imaging
	ultrasound	high resolution	medical and industrial imaging
	thermal conduction	slow, soft field	industrial

40

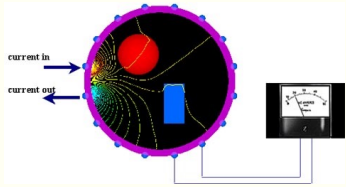
method	sensor element	measured values	typical material properties	typical arrangement	typical materials
ECT	capacitive plates	capacitance C	dielectric permittivity of 1–100, low conductivity $< 10^{-1}\,S^{-1}m^{-1}$		oil, water, powder, polymer, gases
EIT (or ERT)	electrodes	impedance/resistance Z/R	wide conductivity and permittivity range		water, saline, biological tissues, geophysical materials
EMT (or MIT)	coils	self/mutual inductance L/M	wide conductivity and permeability range		metals, some mineral, magnetic materials, conductive water

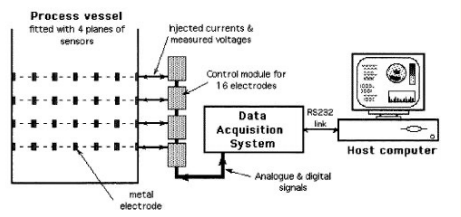
41

ERT(ELECTRICAL RESISTANCE TOMOGRAPHY)

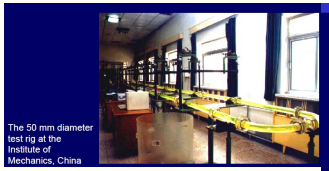
42

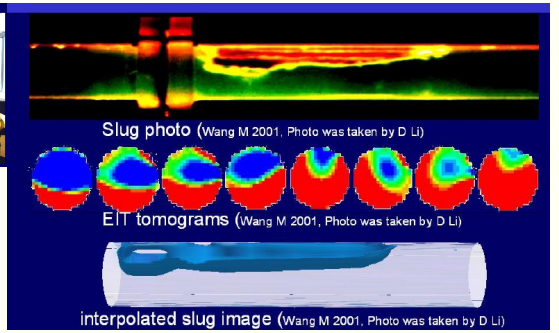
ERT(Electrical Resistance Tomography)





43



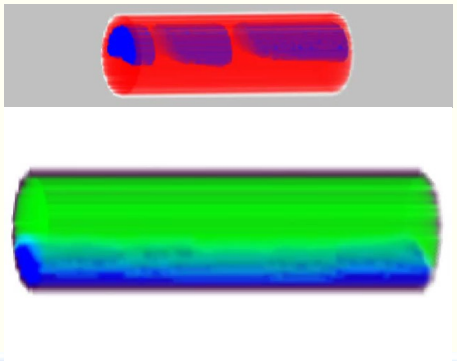


44



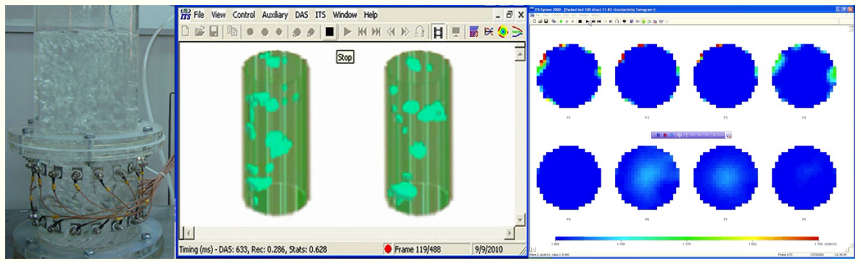
http://cse.zju.edu.cn

45



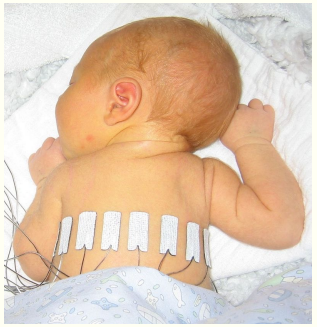
http://cse.zju.edu.cn

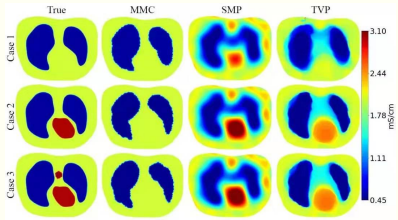
46



http://cse.zju.edu.cn

47





http://cse.zju.edu.cn

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▪应用



讨论：请给出其测量原理和方法。

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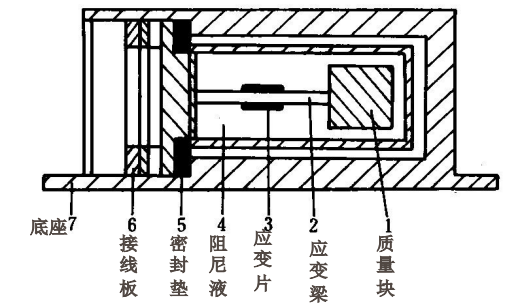


图 2-16　应变式加速度传感器

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▪金属热电阻

- 材料要求?
  - 电阻随温度变化保持单值函数，最好是线性关系；
  - 电阻温度系数大，温度增加时，其电阻值有明显增大；
  - 较高的电阻率，以减小尺寸、减小热惯性；
  - 物理和化学性能稳定，不易被介质腐蚀；
  - 易于得到高纯物质，复现性好，价格便宜。

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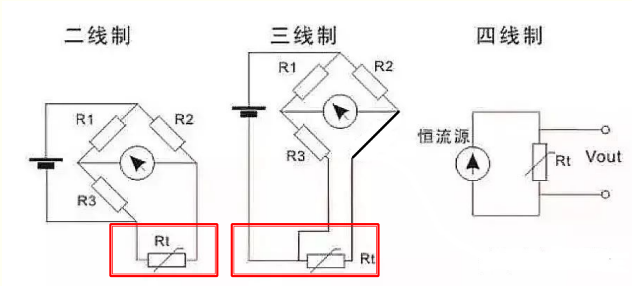
▪**使用注意事项:**
- ▪自热误差
- ▪引线电阻的影响

53

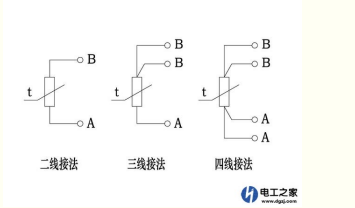
▪**讨论:**
- ▪**如何实现金属热电阻的准确测量?**

54

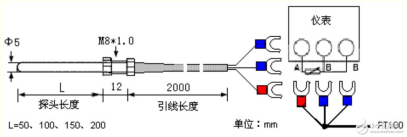
▪**三线制?**

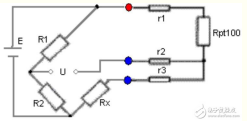


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Pt_{100}

电桥电源：10V

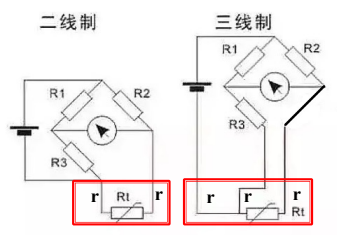
$R_1 = R_2 = 100\Omega$

$R_3 = 100\Omega$

引线电阻$r = 5\Omega$

被测温度300℃

两线制接法引起的相对测量误差?



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金属热电阻? 半导体热敏电阻?

▪金属热电阻

- 正温度系数, 0.4% ~ 0.6% /℃
- 一致性好（可互换）
- 测量范围宽
- 采用三线制接法可减小引线电阻影响

▪热敏电阻

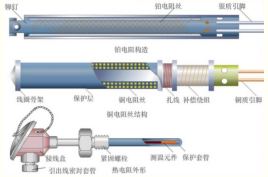
- 负温度系数, -2% ~ 6%/℃
- 一致性差
- 测量温度范围较窄
- 测量电路简单
- 热惯性小（体积小）
- 灵敏度高（电阻温度系数大，是金属热电阻的十几倍）

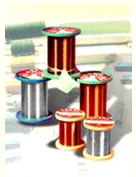
58

▪金属热电阻的选择

- R_0（$t = 0$℃的R）的大小
  - 铂： $R_0 = 10\Omega$、100Ω、$1000\ \Omega$
  - 铜：$R_0 = 50\Omega$和$R_0 = 100\Omega$
- 电阻丝的粗细
  - 铂电阻 $d = 0.03 \sim 0.07$mm
  - 铜电阻 $d = 0.1$mm

讨论：R_0如何选择?



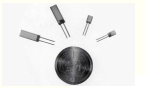


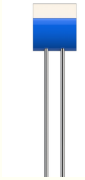
59

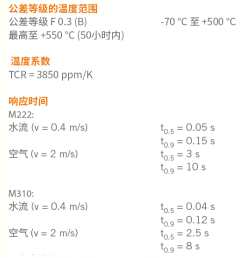
▪如果需要金属热电阻实现快速测量，有什么办法?

- 薄膜热电阻（铂）

- 体积小，尺寸可达：宽度为3μm，长度1.6mm，厚度为0.6mm;
- 响应快，在0.5τ 的条件下，水流速V=0.2m/s 时为0.05s，空气流速V=1m/s 时为4s;
- 一致性好;
- 抗震、抗振动等机械性能好;
- 精度高;
- 测温范围宽;
- 结构多样化;
- 长期稳定性好;
- 标称阻值多从 Pt6.8 到Pt10k







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